

```
6      c=b*23;
7      d=a*b/c*a;
8      printf("Hello %d\n",d);
9      return 0;
10 }
```

(gdb) b 5

Breakpoint 1 at 0x1165: file showBreak.c, line 5.

(gdb) info breakpoints

Num	Type	Disp	Enb	Address	What
1	breakpoint	keep y		0x0000000000001165	in main at showBreak.c:5

(gdb) disassemble main

Dump of assembler code for function main:

```
0x0000000000001149 <+0>:      endbr64
0x000000000000114d <+4>:      push    %rbp
0x000000000000114e <+5>:      mov     %rsp,%rbp
0x0000000000001151 <+8>:      lea     0x00000000(%rip),%rax      # 0x2004
0x0000000000001158 <+15>:     mov     %rax,%rdi
0x000000000000115b <+18>:     call   0x00000000<puts@plt>
0x0000000000001160 <+23>:     mov     $0x0,%eax
0x0000000000001165 <+28>:     pop     %rbp
0x0000000000001166 <+29>:     ret
```

End of assembler dump.

(gdb) b *0x000000000000115b

Breakpoint 2 at 0x115b: file showBreak.c, line 3.

(gdb) info breakpoints

Num	Type	Disp	Enb	Address	What
1	breakpoint	keep y		0x0000000000001165	in main at showBreak.c:5
2	breakpoint	keep y		0x000000000000115b	in main at showBreak.c:3

(gdb) r

```

0x00000000000001151 <+8>: mov %rsp, %rbp
0x00000000000001158 <+15>: lea 0x0000000000000000(%rip), %rax # 0x2004
0x0000000000000115b <+18>: mov %rax, %rdi
0x00000000000001160 <+23>: call 0x1050 <puts@plt>
0x00000000000001165 <+28>: mov $0x0, %eax
0x00000000000001166 <+29>: pop %rbp
ret

```

End of assembler dump.

(gdb) b *0x0000000000000115b

Breakpoint 2 at 0x115b: file showBreak.c, line 3.

(gdb) info breakpoints

Num	Type	Disp	Enb	Address	What
1	breakpoint	keep y		0x00000000000001165	in main at showBreak.c:5
2	breakpoint	keep y		0x0000000000000115b	in main at showBreak.c:3

(gdb) run

Starting program: /home/kolin/col7001/concurrency/a.out

Warning:

Cannot insert breakpoint 2.

Cannot access memory at address 0x115b

(gdb) x/10xb \$rip

```

0x7ffff7fe3290: 0x48 0x89 0xe7 0xe8 0x98 0xd 0x00 0x00
0x7ffff7fe3298: 0x49 0x89

```

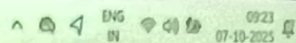
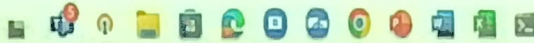
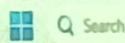
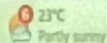
(gdb) x/10xb \$rip-10

```

0x7ffff7fe3286: 0x0f 0x1f 0x84 0x00 0x00 0x00 0x00 0x00
0x7ffff7fe328e: 0x66 0x90

```

(gdb)



Multiprocessor Types

- Loosely coupled multiprocessors
 - No shared global memory address space
 - Multicomputer network
 - Network-based multiprocessors
 - Usually programmed via message passing
 - Explicit calls (send, receive) for communication
- Tightly coupled multiprocessors
 - Shared global memory address space
 - Traditional multiprocessing: symmetric multiprocessing (SMP)
 - Existing multi-core processors, multithreaded processors
 - Programming model similar to uniprocessors (i.e., multitasking uniprocessor) except
 - Operations on shared data require synchronization

Main Design Issues in Tightly-Coupled MP

- Shared memory synchronization
 - How to handle locks, atomic operations
- Cache coherence
 - How to ensure correct operation in the presence of private caches keeping the same memory address cached
- Memory consistency: Ordering of all memory operations
 - What should the programmer expect the hardware to provide?
- Shared resource management
- Communication: Interconnects

Main Programming Issues in Tightly-Coupled MP

- Load imbalance
 - How to partition a single task into multiple tasks
- Synchronization
 - How to synchronize (efficiently) between tasks
 - How to communicate between tasks
 - Locks, barriers, pipeline stages, condition variables, semaphores, atomic operations, ...
- Contention
- Maximizing parallelism
- Ensuring correct operation while optimizing for performance

Hardware-based Multithreading

- Coarse grained
 - Quantum based
 - Event based (switch-on-event multithreading), e.g., switch on L3 miss
- Fine grained
 - Cycle by cycle
 - Thornton, "CDC 6600: Design of a Computer," 1970.
 - Burton Smith, "A pipelined, shared resource MIMD computer," ICPP 1978.
- Simultaneous
 - Can dispatch instructions from multiple threads at the same time
 - Good for improving execution unit utilization

Parallel Speedup Example

- $a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$
- Assume given inputs: x and each a_i
- Assume each operation 1 cycle, no communication cost, each op can be executed in a different processor
- How fast is this with a single processor?

Parallel Speedup Example

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- Assume given inputs: x and each a_i
- Assume each operation 1 cycle, no communication cost, each op can be executed in a different processor
- How fast is this with a single processor?
 - Assume no pipelining or concurrent execution of instructions
- How fast is this with 3 processors?

Superlinear Speedup

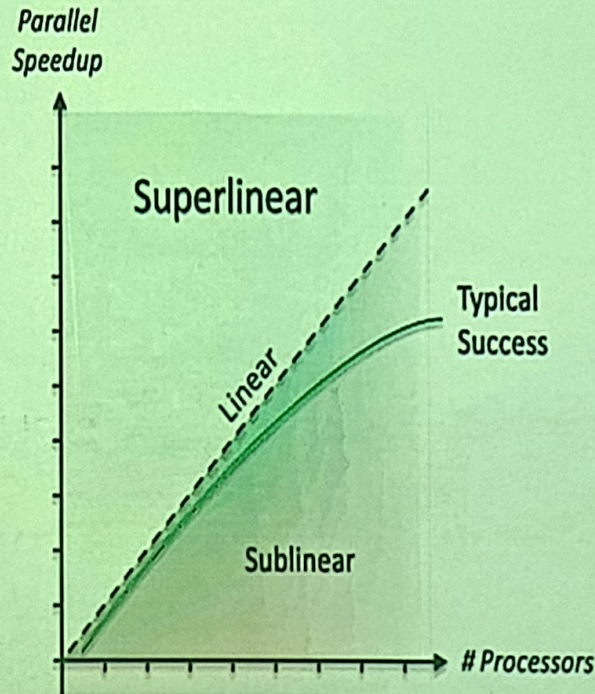
- Can speedup be greater than P with P processing elements?

- Unfair comparisons

Compare best parallel algorithm to wimpy serial algorithm $\rightarrow$ unfair

- Cache/memory effects

More processors $\rightarrow$
more cache or memory $\rightarrow$



Utilization, Redundancy, Efficiency

- Traditional metrics
 - Assume all P processors are tied up for parallel computation
- Utilization: How much processing capability is used
 - $U = (\# \text{ Operations in parallel version}) / (\text{processors} \times \text{Time})$
- Redundancy: how much extra work is done with parallel processing
 - $R = (\# \text{ of operations in parallel version}) / (\# \text{ operations in best single processor algorithm version})$
- Efficiency
 - $E = (\text{Time with 1 processor}) / (\text{processors} \times \text{Time with P processors})$

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Caveats of Parallelism (II)

- Amdahl's Law
 - f : Parallelizable fraction of a program
 - N : Number of processors

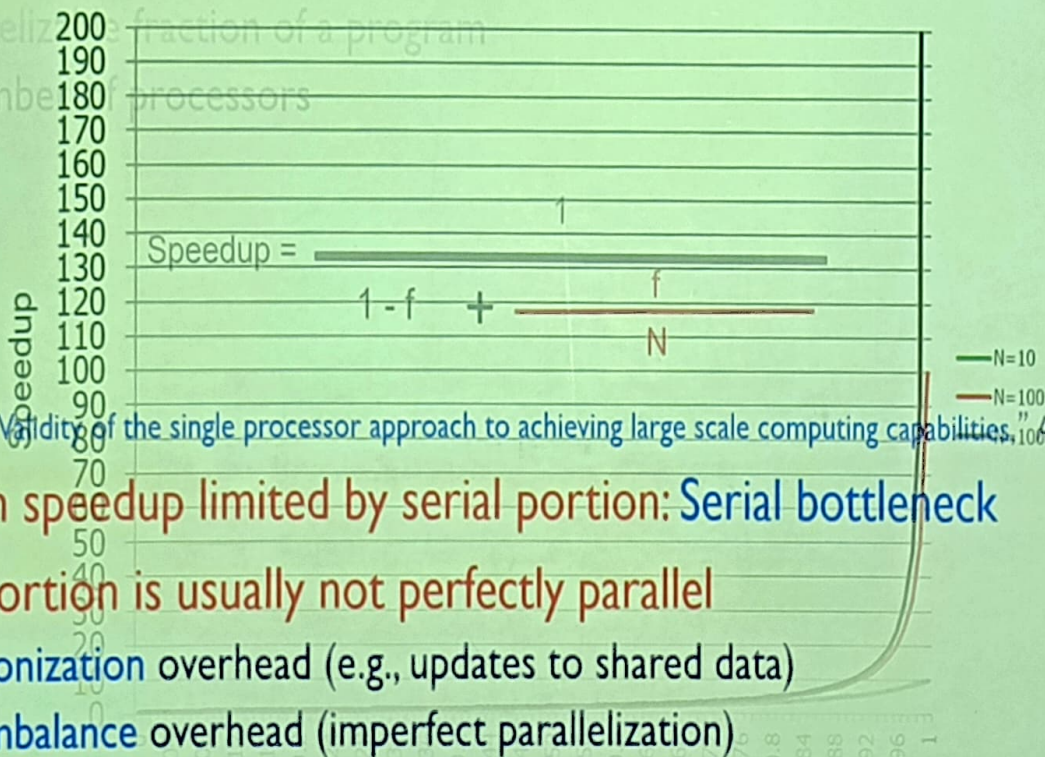
$$\text{Speedup} = \frac{1}{1 - f + \frac{f}{N}}$$

- Amdahl, "Validity of the single processor approach to achieving large scale computing capabilities," AFIPS 1967.
- Maximum speedup limited by serial portion: Serial bottleneck
- Parallel portion is usually not perfectly parallel
 - Synchronization overhead (e.g., updates to shared data)
 - Load imbalance overhead (imperfect parallelization)
 - Resource sharing overhead (contention among N processors)

Sequential Bottleneck

- Amdahl's Law

- f: Parallelizable fraction of a program
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What is the Maximum Speedup?

- f = fraction of computation (algorithm) that is serial and cannot be parallelized
 - Data setup
 - Reading/writing to a single disk file
- $t_s = ft_s + (1-f) t_s$
= serial portion + parallelizable portion
- $t_p = ft_s + ((1-f) t_s)/n$
- $S(n) = t_s/(ft_s + ((1-f) t_s)/n)$
= $n/(1 + (n-1)f)$ <- Amdahl's Law

Limit as $n \rightarrow \infty = 1/f$

Example of Amdahl's Law

- Suppose that a calculation has a 4% serial portion, what is the limit of speedup on 16 processors?
 - $16 / (1 + (16 - 1) * .04) = 10$
 - What is the maximum speedup?
 - $1 / 0.04 = 25$

More to think about ...

- Amdahl's law works on a *fixed* problem size
 - This is reasonable if your only goal is to solve a problem faster.
 - What if you also want to solve a larger problem?
 - Gustafson's Law (Scaled Speedup)

Gustafson's Law

- Fix execution of on a single processor as
 - $s + p = \text{serial part} + \text{parallelizable part} = 1$
- $S(n) = (s + p)/(s + p/n)$
 $= 1/(s + (1 - s)/n) = \text{Amdahl's law}$
- Now let, $1 = s + \pi = \text{execution time on a parallel computer, with } \pi = \text{parallel part.}$
 - $S_s(n) = (s + \pi n)/(s + \pi) = n + (1-n)s$

More on Gustafson's Law

- Derived by fixing the parallel execution time
 - Amdahl fixed the problem size
 - fixed serial execution time
 - For many practical situations, Gustafson's law makes more sense
 - Have a bigger computer, solve a bigger problem.
- Amdahl's law turns out to be too conservative for high-performance computing.

Example questions

- Given a (scaled) speed up of 20 on 32 processors, what is the serial fraction from Amdahl's law?, From Gustafson's Law?
- A program attains 89% efficiency with a serial fraction of 2%. Approximately how many processors are being used according to Amdahl's law?

Efficiency

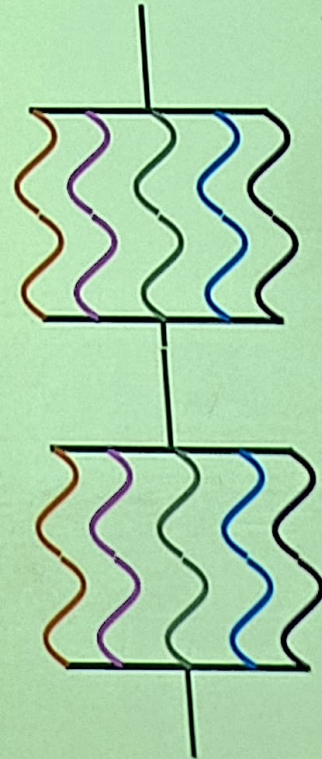
- $E(n) = S(n)/n * 100\%$
- A program with linear speedup is 100% efficient.

Example questions

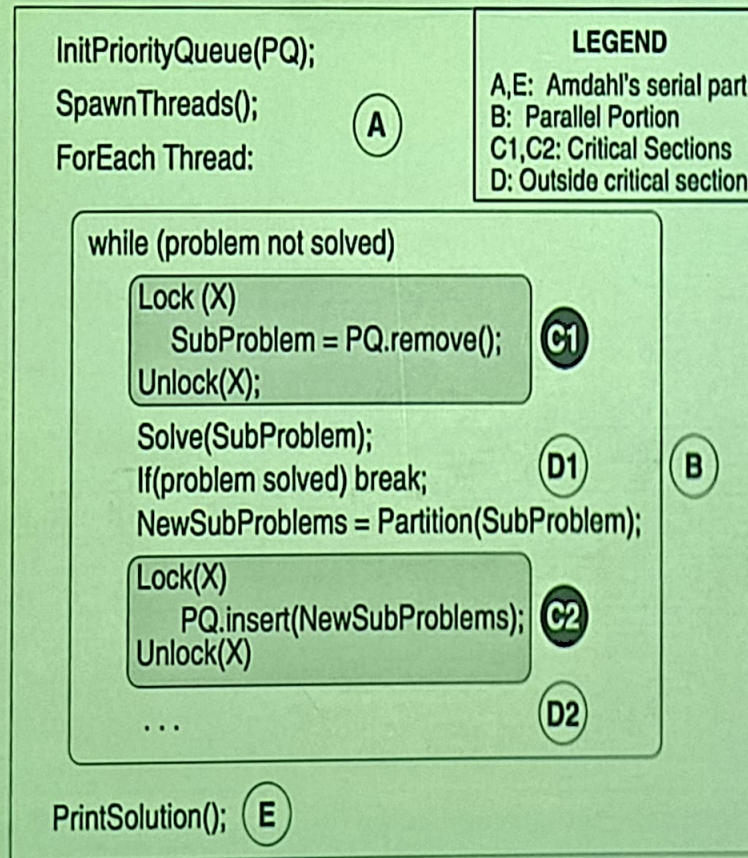
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Why the Sequential Bottleneck?

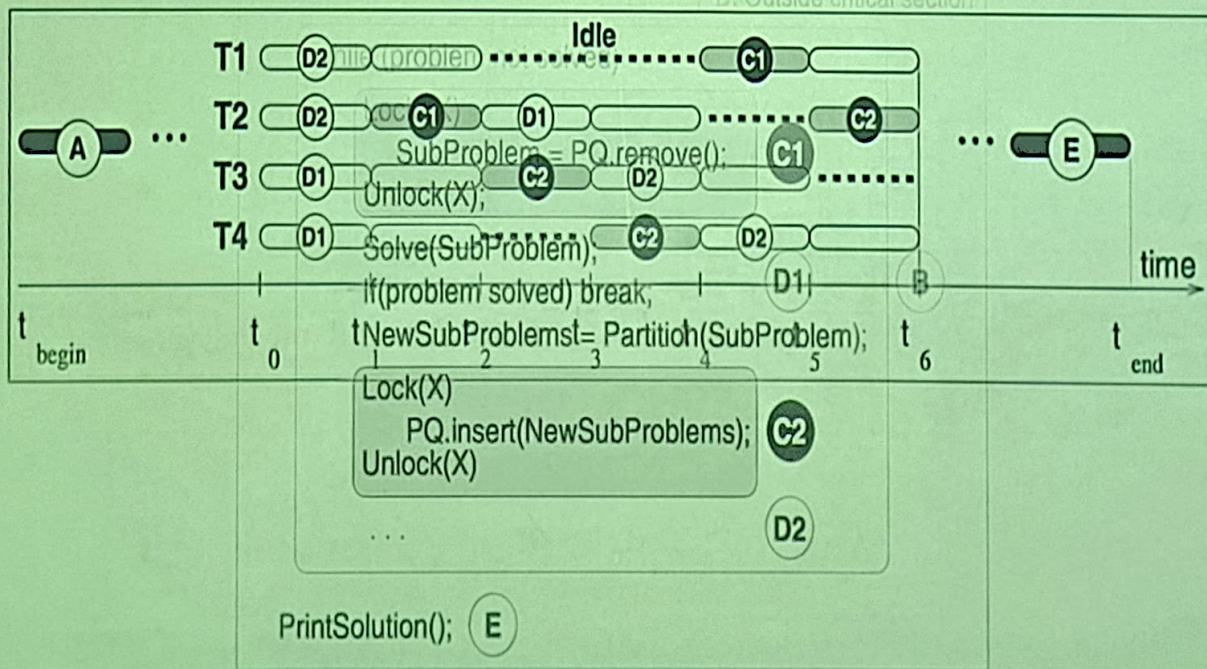
- Parallel machines have the sequential bottleneck
- Main cause: **Non-parallelizable** operations on data (e.g. non-parallelizable loops)
for (i = 0 ; i < N; i++)
 $A[i] = (A[i] + A[i-1]) / 2$
- There are other causes as well:
 - Single thread prepares data and spawns parallel tasks (usually sequential)



Another Example of Sequential Bottleneck (I)



Another Example of Sequential Bottleneck (II)



Bottlenecks in Parallel Portion

- **Synchronization:** Operations manipulating shared data cannot be parallelized
 - Locks, mutual exclusion, barrier synchronization
 - **Communication:** Tasks may need values from each other
 - Causes thread serialization when shared data is contended
- **Load Imbalance:** Parallel tasks may have different lengths
 - Due to imperfect parallelization or microarchitectural effects
 - Reduces speedup in parallel portion
- **Resource Contention:** Parallel tasks can share hardware resources, delaying each other
 - Replicating all resources (e.g., memory) expensive

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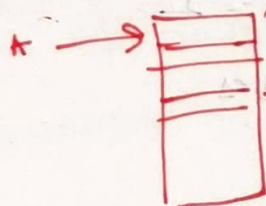
7-Oct-2025 Marking scheme

How do you get an executable?

→ Steps X No marks

Expected — gcc
c pp
ccl
in
ld

- Given int - takes 4 Bytes show bytes
- loop not allowed to get content / printing not allowed
- understand what is Memory



* p = &n

use a unsigned / char pointer

Why every C prog has main?

- is it really start point?

- Some code / frame there who does that

— startup is start point

Don't believe a book that says main is start point

k(0)
k(1)
p(2)
k(3)

Does printf work?

How

→ 2 stacks

pass from 1 stack to another

printf is writing to his stack

How GDB works?

— can not work — without

interrupt3

Done via ptrace

(peek and fork)

Modern compiler generate pos independent code
and generally they Randomise stack

apply breakpoint

[Commands & explanation]

b 5

→ breakpoint on line 5

into breakpoints

→ shows all breakpoints in prog.

disassemble main

→ shows Assembly code (optimised)

run

→ Runs code with Break points

x / 10xb \$grip

→ Address of instructions from current instruction pointer

- GDB doesn't show interrupts it adds

- Back to concurrency bugs

(or)

Parallelism can be in form of tasks, instructions

↓
↳ diff tasks

↓
1 to n
 $a_i = a_i + b_i$

Multi processors

communicate via message
passing

↳ loosely coupled

[Don't share memory]

ex
N/w based
processor

↳ tightly coupled

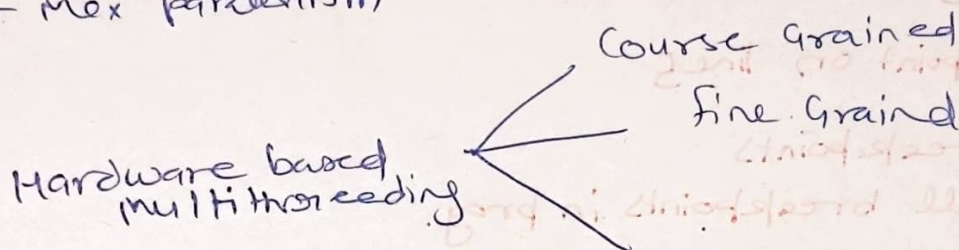
[share memory]

↳ multicore processor

communicate via Synchronisation

Tightly Coupled M.P.

- Synchronisation
- Memory consistency
- contention
- Max parallelism



Parallel speed up Ex

$$a_4 n^4 + a_3 n^3 + a_2 n^2 + a_1 n + a_0$$

Honor's Rule?

- reuse power of n

Do Bracketing

$$\left(\left(\left(a_4 n + a_3 \right) n + a_2 \right) n + a_1 \right) n + a_0$$

8 op

Now

How Do we execute it in parallel

→ 4 processor speed up

It is difficult to beat linear time, p processors will have speedup of p

Superlinear Speedup

→ In theory it's not possible

→ In real machines it's possible

→ count seconds (cache)

+ How far data from processor

→ searching in a 100k integers

↓
100 processors

Speedup

1

vs

100

↑
(more speedup)

Branch and bound

Each of branches can evaluate in parallel condition

for sorting what Algo you use in parallel processing

- Idea change here

- what memory you are working on you must see

- Data transfer is more expensive than computation

$$\text{Utilization} = \frac{\text{\#operations in 11al \#version}}{\text{processors} \times \text{time}}$$

$$\text{Speedup} = \frac{1}{1 - s + \frac{s}{N}} \Leftrightarrow \frac{1}{s + \frac{1-s}{N}}$$

Amdahl's law

You are fundamentally limited by No. of processors

as $s \rightarrow 0$, that fraction becomes 0
bottleneck serial portion

parallel portion is not perfectly parallel

~~Synchronisation~~

Gustafson's law

→ I have a problem, can parallelise in n parts

how much can we speedup

~~other~~ other way to look

— Application has alot of Res, clearly I can get more output

— Diff eq. are essentially Integration

↓
~~fast~~
accuracy goes up, can solve larger problem

Sequential bottleneck

— non parallel part of Data

— Synchronisation

— load imbalance

— Resource Contention